

On the influence of lid materials for flip-chip ball grid array package applications

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ABSTRACT

Autonomous driving is a key vehicle capability for future mobility solutions that relies on the reliability of its high-performance vehicle computer. As for any electronics for automotive applications, thermal management is a crucial point. A very extensively employed approach for improving heat removal is the usage of heat spreaders in the form of a package lid, especially for flip-chip applications. The usage of a lid in flip-chip ball grid array (BGA) packages improves heat removal based on the heat spreading capabilities of the lid, and reduces package warpage, since with a lid the package becomes stiffer. However, adding a lid also increases stresses at solder interconnects. When selecting a material for the lid, the most intuitive criterion is to focus on the material's thermal conductivity. Sometimes overlooked, the coefficient of thermal expansion (CTE) of the lid also plays an important role regarding package warpage and reliability of thermal interface materials (TIM) as well as solder interconnects, such as solder bumps. This work focuses on the influence of different lid materials at package level for flip-chip applications. Consequently, the influence of three alternative materials is investigated and compared to the performance of a standard copper lid. For the quantification of the effects, a validated finite element model of a flip-chip package is presented, with the thermo-mechanical performance of the studied lid materials validated based on package warpage, peel strains and plastic strain energy density at the bumps, as well as on peel and shear strains at the TIM.

1. Introduction

Autonomous driving (AD) is a key vehicle capability for future mobility solutions, contributing to improving traffic efficiency [1,2], emission of greenhouse gases [1,3] and safety [1,2,4]. The central part in an AD vehicle is its electronic control unit (ECU), i.e., a high-performance vehicle computer (HPVC) responsible for performing complex tasks, such as object recognition, vision processing and high-speed data transfer between different ECUs in the vehicle. Hence, a reliable HPVC is paramount.

More than 50% of electronics failure is connected to temperature management [5], and higher temperature ranges induce higher thermo-mechanical stresses caused by the coefficient of thermal expansion (CTE) mismatch between components [6]. In order to improve heat removal, heat spreaders in the form of lid have been extensively employed in the industry for flip-chip applications [7]. The lid is usually attached to the silicon die via a thermal interface material (TIM) and

to the substrate via an adhesive, forming a so-called lidded package or thermally enhanced flip-chip package (TEFP). A cross-section of a lidded package is depicted in Fig. 1.

There are many parameters that influence the reliability of flip-chip packages. The influence of the adhesive material on package reliability has been studied in [7,8]. Package warpage is also a critical point to be addressed and comprehensive studies have been published on package warpage [9–15] and its prediction [16–20].

In the assembly process of flip-chip packages, more specifically after the reflow of solder bumps, the resulting warpage caused by the CTE-mismatch between die and substrate is not yet an issue due to the relaxation of the solder bumps. The warpage starts to be more pronounced and, therefore, to become an issue after the dispensing and curing of the underfill. At this stage, excessive warpage of the package can lead to challenges for board level interconnects, such as solder ball bridging or opening. Influence of underfill selection and implication

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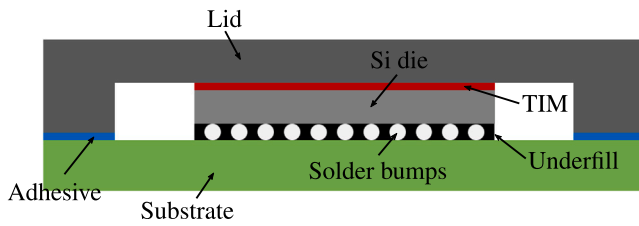


Fig. 1. Cross-section of a typical lidded electronic package.

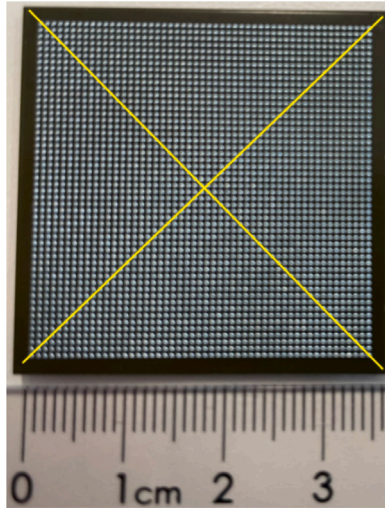


Fig. 2. Diagonals used to measure substrate warpage.

on package reliability, as well as underfill failure modes can be found in [21–27].

To avoid such warpage related issues, different approaches have been proposed to increase the stiffness of the package, *i.e.*, to control warpage, such as encapsulating the die [28], adding a stiffener ring [29, 30], capping the die [31], or employing a two-piece or one-piece lid [30,32]. All approaches have their advantages and disadvantages, for instance, encapsulation reduces flexural strains at the chip during bending for ultra-thin dies [28], but it has low reworkability and process related issues might occur, such as void formation [33]. Using a stiffener ring reduces the total thermal resistance for top-side die cooling, but the warpage control effect is less effective when compared to the approach with lid. Capping the die allows a very precise warpage control, while compromising the heat removal capabilities [31]. For the two-piece lid, the stiffness of the package can be better tuned by selecting different materials for the parts, but the two-piece lid leads to higher package stress when compared to the one-piece variant [32]. The approach with one-piece lid leads to the smallest warpage of the flip-chip package but conversely causes higher stresses [31]. One-piece lids are still very attractive due to factors like processing steps, handling costs, reworkability, *etc.* Hence, finding alternatives that balances warpage control and reduction of stress levels can make the usage of one-piece lid even more attractive.

In this regard, this work focuses on the study of different materials for the one-piece lid (henceforth lid) for flip-chip package applications, *i.e.*, at package level, the influence of three different lid materials is investigated and compared to the performance of a standard copper lid. The selection of these materials was based on their properties, with focus on the CTE and Young's modulus (E), given their influence on the CTE-mismatch and overall stiffness of the package, respectively. By the selection, the goal was to study materials that are commercially available and share only one similar mechanical property, while the others are different, *i.e.*, materials that have similar E but different CTE,

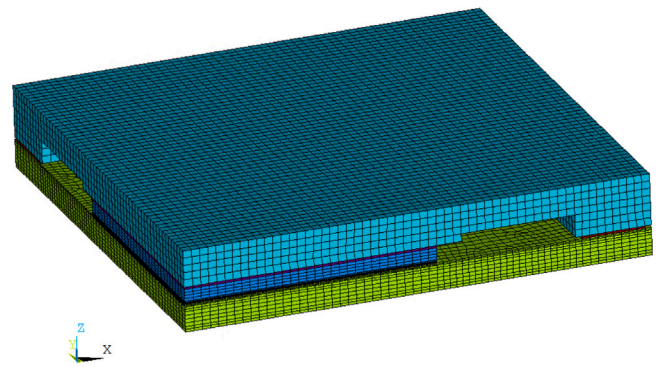


Fig. 3. Flip-chip package FE model.

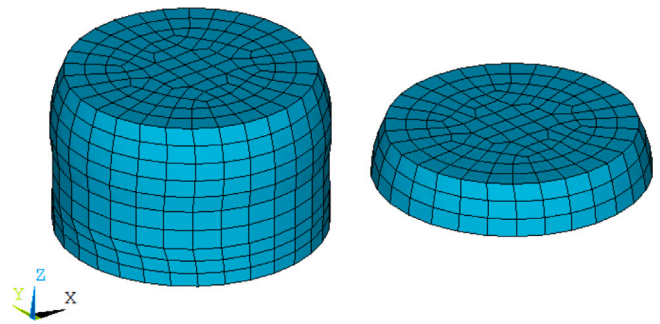


Fig. 4. FE mesh of a solder bump (left) with the elements used for volume averaging (right). The mesh refers to the solder bumps depicted in Fig. 10.

and materials that have similar CTE and different E , and with that to understand the influence of these properties on the thermo-mechanical responses of the flip-chip package. The selected materials for the study are presented in Table 1. In order to quantify the effects of using these materials, coupled thermal and thermo-mechanical numerical analysis with a validated numerical model were conducted, with the thermo-mechanical performance evaluated based on the peel strain and plastic strain energy density at solder interconnects and on the peel and shear strains at the TIM.

2. Numerical model and validation

For the measurements, 12 samples with a 37 mm × 37 mm substrate and a 20 mm × 20 mm die were lidded with the three different materials of interest, namely a ceramic (Mat. A), a ceramic matrix composite (Mat. B), a metal matrix composite (Mat. C) and with the reference material copper (Table 1), resulting in 3 flip-chip lidded packages for each lid material. For the experimental setup, one sample at a time was placed inside an oven with 3 thermocouples of type K for recording the temperature. One thermocouple is placed at the lid, while the other two are located in the oven's chamber. For the measurements, the samples were kept at 25 °C, 80 °C, 110 °C and 125 °C for at least 10 min to guarantee steady-state, which was confirmed by the data from the 3 thermocouples. The projection moiré technique [34,35] was then used to measure the warpage of the substrate. For each sample, the warpage of the substrate was measured along both diagonals, as depicted in Fig. 2.

For the numerical simulations, a three-dimensional (3D) quarter model was prepared for the finite element (FE) analysis. As referenced in Fig. 1 and shown in Fig. 3, the numerical model comprises a Si die attached to a substrate via solder connections (bumps) and underfill. The FE mesh of a solder bump is shown in Fig. 4. The lid is attached to the substrate via an adhesive and to the die by a TIM. The temperature dependence of the material properties of underfill, adhesive, solder,

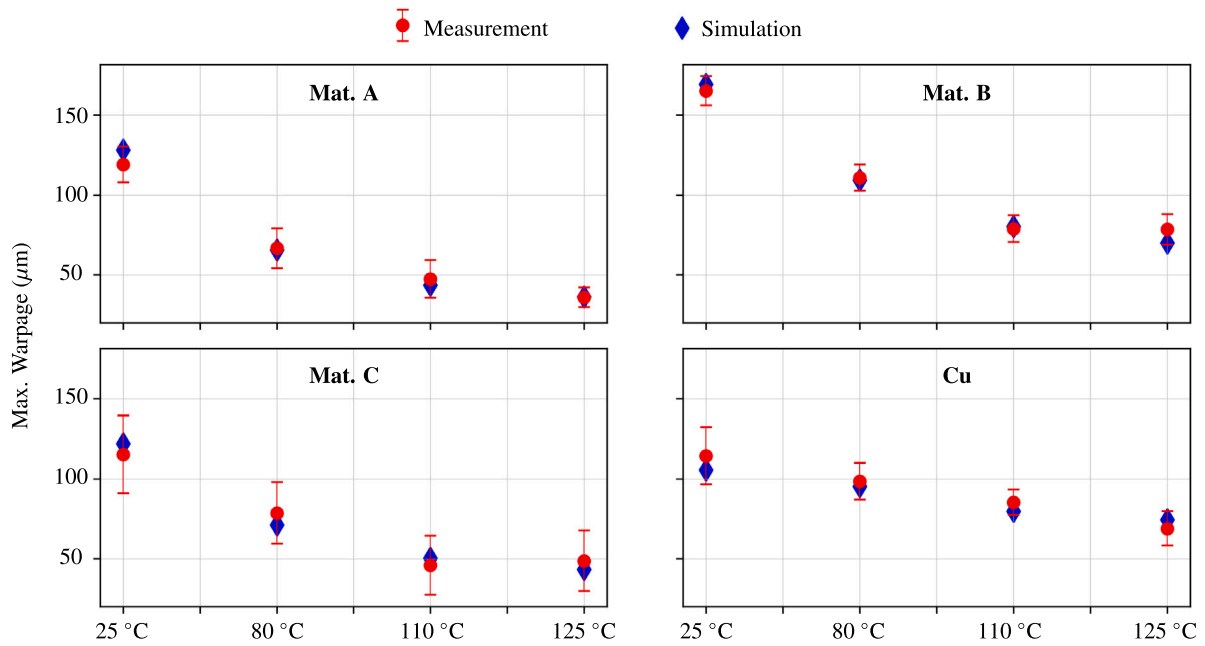


Fig. 5. Comparison between experimental and simulation results.

Table 1

Thermo-mechanical properties of the materials used for the package lid.

Material/Property	E (GPa)	CTE ($10^{-6}/K$)	λ (W/m/K)
Mat. A	320	3.1	160
Mat. B	120	8.3	100
Mat. C	220	7.7	200
Cu [36]	128	17	392

Table 2

Material properties and component thickness used in simulation.

Component	$E_x E_y E_z$ (GPa)	CTE _x CTE _y CTE _z ($10^{-6}/K$)	$\lambda_x \lambda_y \lambda_z$ (W/m/K)	T_g (°C)	t (mm)
Si Die	170 170 130	2.8	120	–	0.525
Underfill	5	60	0.40	130	0.050
Adhesive	0.12	92	3.20	–45	0.115
Solder	50	20	60	–	0.050
Substrate	35 35 15	13 13 30	10 10 6	220	1.20
TIM	0.12	92	3.20	–45	0.090

substrate and TIM are considered in the simulations, and their values at room temperature are given in Table 1 and 2. The glass transition temperature (T_g) of underfill, adhesive, substrate and TIM, as well as the thickness t of the components are also given in Table 2. The lid has a thickness of 1.45 mm. The value of the properties used in the simulation are from characterization data. Except for the bumps, which were modeled with the Chaboche nonlinear material model [37,38], all other components are assumed to follow a linear elastic material model.

The numerical model was validated based on the warpage measurements at the four temperatures mentioned above and the comparison between experimental and numerical results can be seen in Fig. 5. The bars in the experimental data represent a confidence interval of 95% for the measurements.

3. Finite element analysis

In order to compare the performance of the different lid materials, passive temperature cycles were simulated using the commercially available FE software Ansys [39]. Starting at the temperature at which the lid was attached, *i.e.*, 110 °C, the model is cooled down to 25 °C, from where the passive temperature cycle starts: with a constant rate

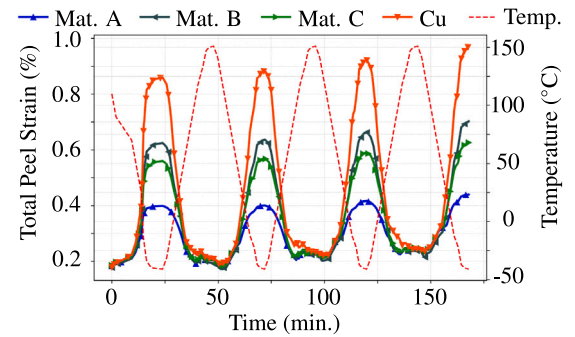


Fig. 6. Peel strain at the solder bumps and temperature load profile used in the FE analysis.

of 10 °C/min, the model is cooled down to –40 °C followed by a dwell time of 5 min at the same temperature. Afterwards, the model is heated up to 150 °C, where also a 5 min dwell time is applied, finishing the first temperature cycle. A total of 3 temperature cycles was simulated. The temperature load profile used in the FE analysis can be seen in Fig. 6. Symmetry boundary condition is also applied at the symmetry planes of the quarter model. The stresses at the package that arise during the assembly process, *i.e.* at time zero in the FEA, are taken into account by means of the stress-free temperature of each component. The following stress-free temperatures were used: 110 °C for the lid, adhesive and TIM; 220 °C for the Si chip, substrate and solder; and 150 °C for the underfill.

Based on their influence on the reliability of flip-chip packages [40–46], the following thermo-mechanical responses are used to quantify the influence of the different lid materials under study: package warpage; peel strain and plastic strain energy density at the first level interconnect, *i.e.*, at the solder bumps; and TIM peel and shear strains.

The evaluation of the warpage of the package is of relevance since it is connected to challenges regarding board level interconnects. The peel strain and the plastic strain energy density at the bumps can be directly associated with crack formation and propagation and are used for evaluating the reliability of solder interconnects [44,47–49].

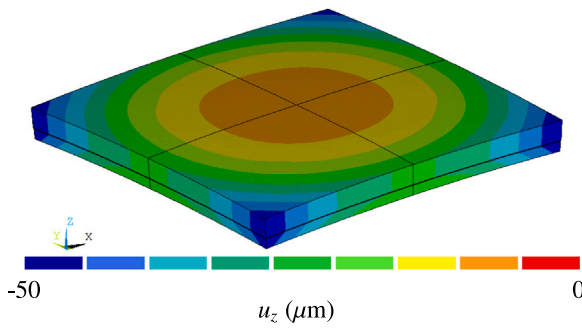


Fig. 7. Flip-chip package warpage at $-40\text{ }^{\circ}\text{C}$ with Cu lid.

For the TIM, the peel strain is relevant due to its impact on thermal performance degradation due to cohesive and adhesive failures and pump-out [50–53], while the shear strain is related to interfacial delamination [54,55].

4. Results

4.1. Flip-chip package warpage

All lid materials studied result in a convex shaped warpage of the package for all temperatures simulated. A typical deformation mode can be seen in Fig. 7 for the package with Cu lid. Warpage results along the package's diagonal for the quarter model for all lid materials at four different temperatures are depicted in Fig. 8. It can be seen that, except for the package with Cu lid, all other lid materials exhibit the highest warpage at $-40\text{ }^{\circ}\text{C}$ whereas the lowest at $150\text{ }^{\circ}\text{C}$. The package with Cu lid exhibits the highest warpage at $25\text{ }^{\circ}\text{C}$ whereas the lowest at $-40\text{ }^{\circ}\text{C}$.

Considering the difference between the maximum and minimum warpage that the flip-chip package exhibits during the temperature cycles, i.e., a delta warpage (Δw), one has to note that the package with Cu lid displays the minimum $\Delta w = 54.68\text{ }\mu\text{m}$, while the maximum $\Delta w = 223.2\text{ }\mu\text{m}$ results from the package with lid Mat. A, followed by Mat. B and Mat. C with $\Delta w = 174.6\text{ }\mu\text{m}$ and $\Delta w = 161.8\text{ }\mu\text{m}$, respectively.

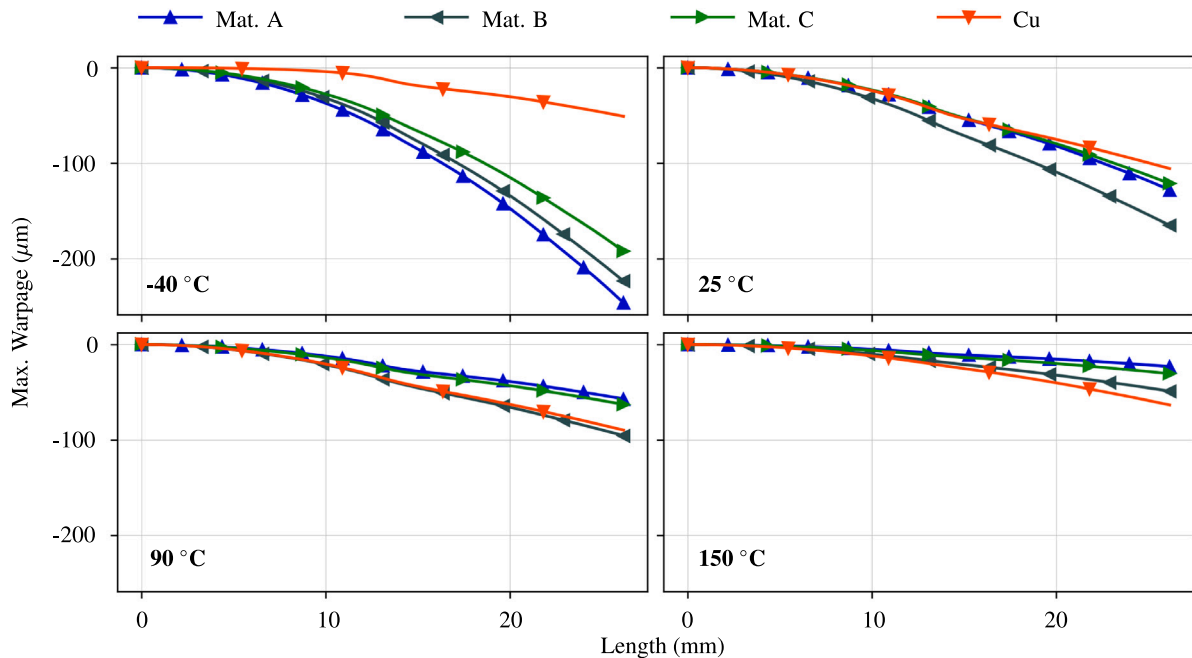


Fig. 8. Warpage results for all lid materials.

Interestingly, the results for the Δw have a negative correlation with the materials' CTE. The same is not observed for the Young's modulus.

In order to shed some light towards the influence of the lid's Young's modulus and CTE on the warpage, a virtual design of experiment (vDoE) was performed using the FE model (Fig. 3). For the vDoE, only E and CTE of the lid are varied and the resulting response surface for the warpage is shown in Fig. 9. It can be seen through the steepness of the surface along the CTE-axis that the CTE of the lid is the main property influencing the warpage. For materials with higher E ($\approx 400\text{ GPa}$), the influence of the CTE of the lid material is minimized, with its influence becoming more pronounced as the E of the lid reduces. Lower E materials ($\approx 100\text{ GPa}$) contribute to a higher influence of the lid's CTE on the package warpage. Similar results concerning the influence of the lid's CTE on the warpage were also published in [56].

4.2. Peel strain and plastic strain energy density at bumps

To assess the influence of the different lid materials with respect to reliability of bumps, the maximum peel strain and the plastic strain density are considered. For the evaluation, four solder bumps located at the edge of the silicon die shadow are used, as shown in Fig. 10. To address the issue concerning edge singularity at the interface solder bump/copper pad, the volume average technique [49] is applied over a $25\text{ }\mu\text{m}$ thick layer on the top side of the solder bumps to evaluate the responses, as shown on the right side of Fig. 4.

It can be seen in Fig. 6 that the maximum peel strains at the bumps at lower temperatures are reduced in comparison to Cu when Mat. A, B and C are used for the lid, whereas at higher temperatures the peel strains are roughly at the same level for all materials.

A positive correlation between the peel strain and CTE can be observed, i.e., at lower temperatures, the maximum peel strain at the bumps is directly proportional to the CTE of the materials, with the lid with the smallest CTE resulting in the smallest maximum peel strain and the material with the highest CTE leading to the highest maximum peel strain at the bumps. This positive correlation is confirmed by the slope of the response surface along the CTE-axis in Fig. 11.

As observed for the peel strain, the resulting plastic strain energy density at the solder bumps in Fig. 12 has also a positive correlation to the CTE of the lid. The package with Cu lid was used for the

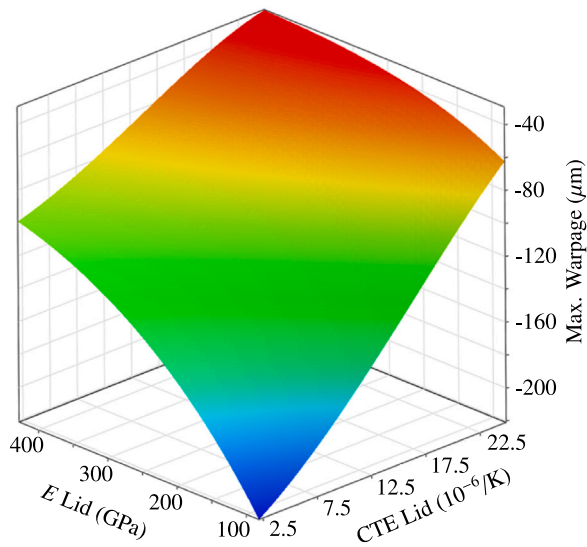


Fig. 9. Response surface for the maximum package warpage.

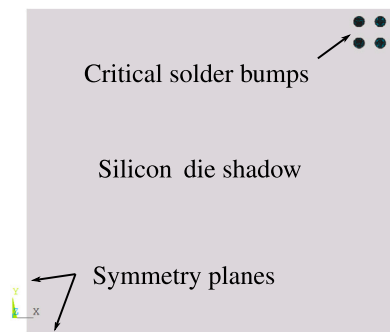


Fig. 10. Location of the solder bumps used in the evaluation.

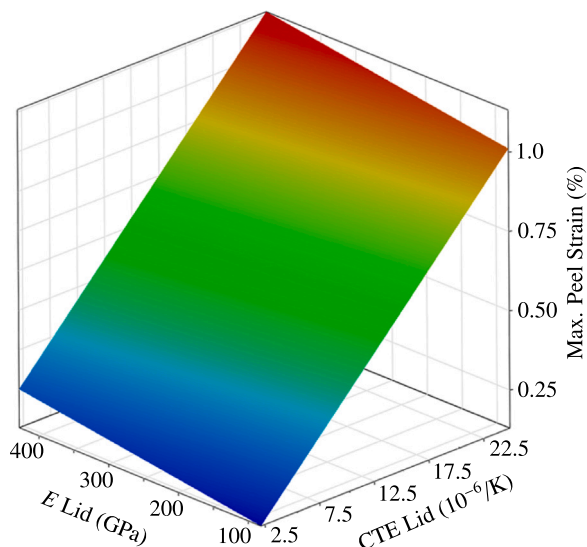


Fig. 11. Response surface for the maximum peel strain at solder bumps.

normalization of the results. After three passive temperature cycles, the packages with lids with Mat. B, C and A result in a reduction of the plastic strain energy density of 41%, 28% and 32%, respectively, in comparison to the results of the package with the reference lid material

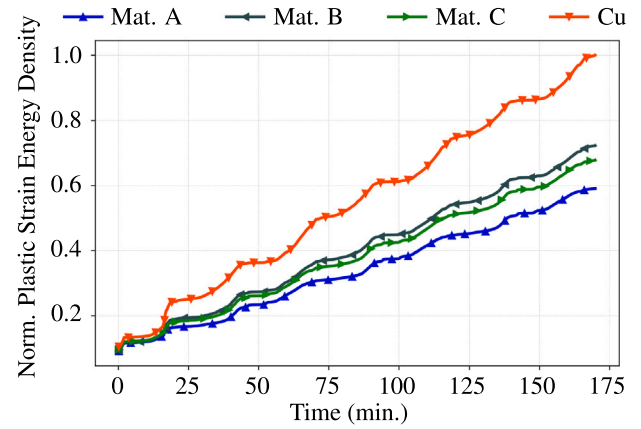


Fig. 12. Normalized plastic strain energy density at the solder bumps.

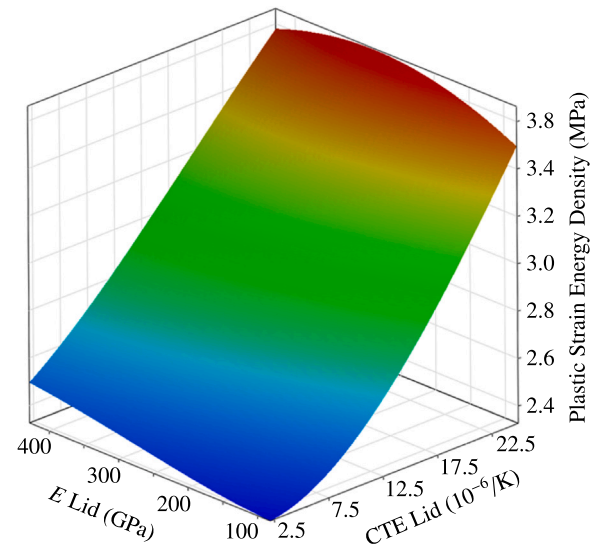


Fig. 13. Response surface for the plastic strain energy density at solder bumps.

Cu. The effect of the plastic strain energy density can be seen in Fig. 6, where the maximum and minimum total peel strain values increase every temperature cycle.

The results of the vDoE for the plastic strain energy density at the bumps are plotted in Fig. 13, where the positive correlation between CTE and plastic strain energy density can be seen. Although the Young's modulus also accounts for variation on the plastic strain energy density, its influence is minor specially when compared to the influence of the CTE.

4.3. Peel and shear strains at TIM

The contour plots from the FE simulations for the peel and shear strains at the TIM are shown in Fig. 14.

For all lid materials, compressive strain is predominant at -40°C . In Fig. 14 it can be seen that Mat. A and B exert a more homogeneous compression at the TIM, while a certain degree of inhomogeneity at the edges starts to appear with the lid Mat. C and Cu. This observation can be related to the warpage of the package at -40°C (Fig. 8), where an inflection point in the curve can be seen where the die shadow finishes. The level of compression that the TIM undergoes can be related to the CTE of the lid materials. The highest level of compression is observed for the lid Mat. A, which has the lowest CTE, while the lowest level

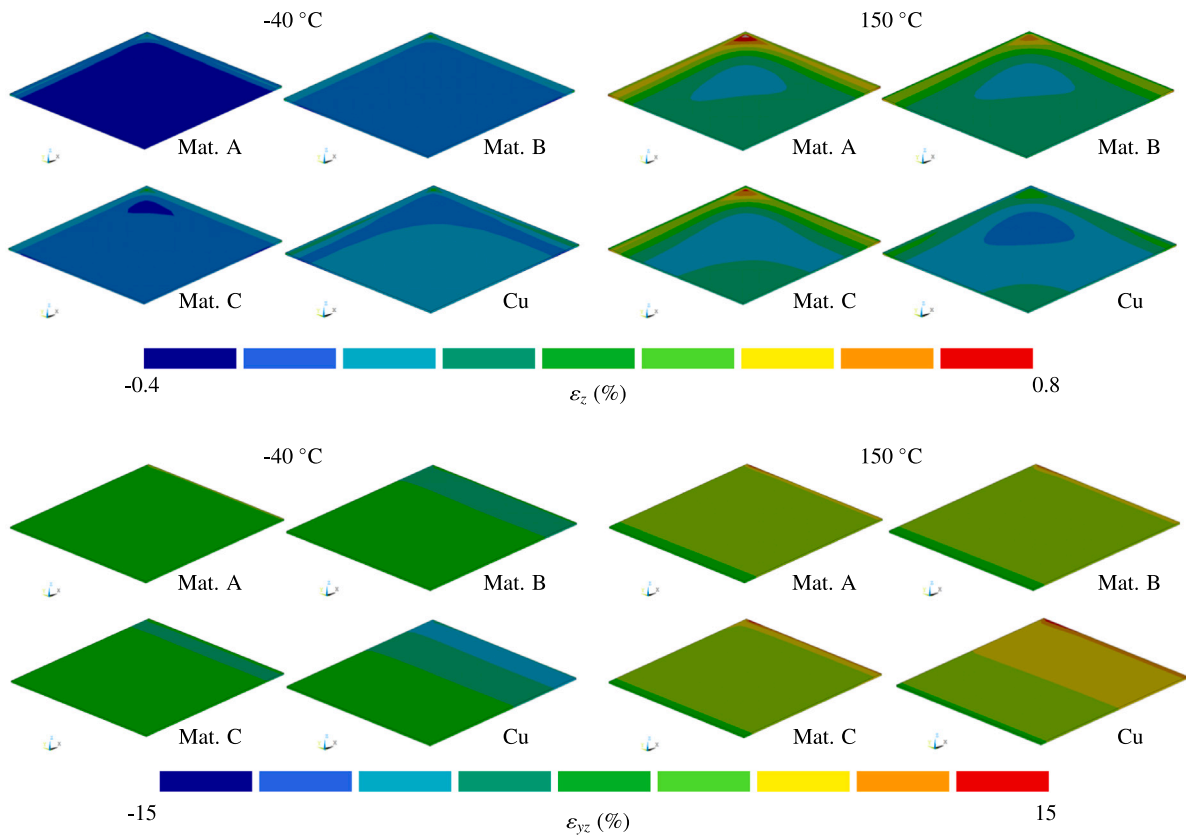


Fig. 14. TIM peel strain at -40 °C and 150 °C at top left and right, respectively and shear strain at -40 °C and 150 °C at bottom left and right, respectively.

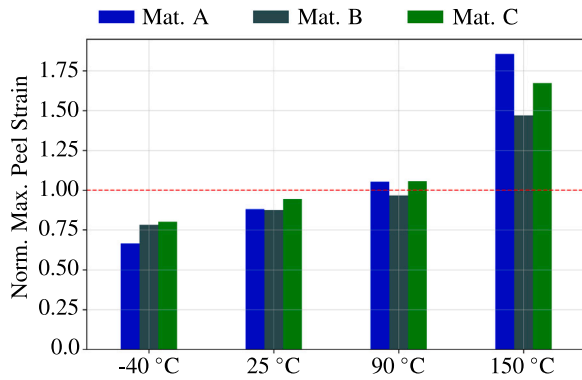


Fig. 15. TIM normalized maximum peel strain.

of compression is observed for the Cu lid, which has the highest CTE among the materials studied.

At 150 °C, a mixture of compression and tension can be observed, with tension dominating for all materials except for Cu, which still displays a considerable amount of compression. Once more, this can be related to the package warpage, since at 150 °C the package with Cu lid exhibits the highest warpage.

Fig. 15 shows the normalized maximum peel strain at the TIM, with the normalization done with respect to Cu. With Mat. A, B and C, the maximum peel strain at the TIM is reduced from -40 °C to up to 90 °C, where Mat. A and C starts to result in higher peel strain in comparison to Cu. At 150 °C, Cu results in the lowest maximum peel strain.

Accordingly, the peel strain state at the TIM can be directly associated with the amount of warpage the package undergoes and, therefore, to the CTE of the lid material, i.e., for more pronounced warpage (>100 μm) a compression dominated strain state is expected. For less

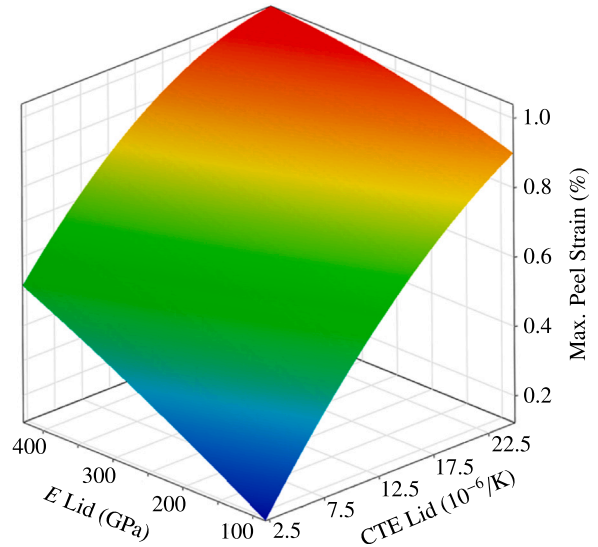


Fig. 16. Response surface for the maximum peel strain at TIM.

pronounced warpage (<100 μm), a tension dominated peel strain state is expected for materials with low CTE ($<9 \times 10^{-6} \text{ K}^{-1}$), but a trend can be noticed towards a more compression dominated peel strain state as the CTE of the lid material increases.

Results from the vDoE for the maximum peel strain at the TIM are shown in Fig. 16. It can be seen that the influence of the CTE of the lid is stronger for softer materials than for harder materials. A similar observation can be made for the Young's modulus, where its influence is stronger for lower CTE materials than for materials with higher CTE.

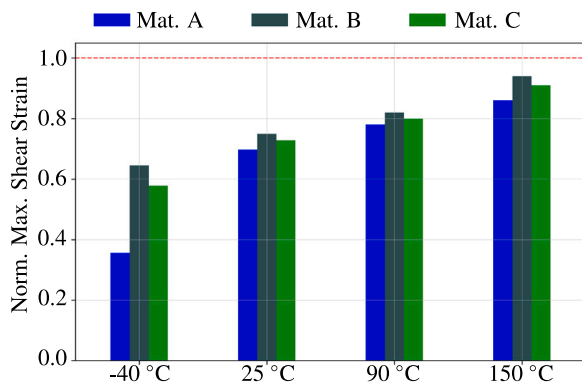


Fig. 17. TIM normalized maximum shear strain.

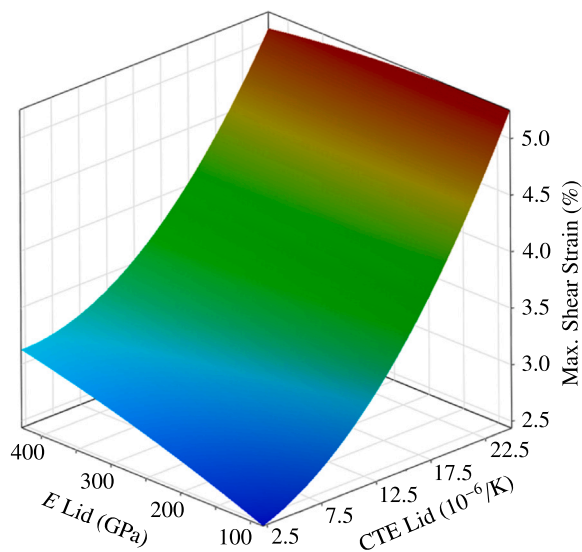


Fig. 18. Response surface for the maximum shear strain at TIM.

In Fig. 14, in the interest of succinctness, only the yz -shear strains are plotted. Ignoring the truncation error from the numerical simulation, the absolute values of the results for the xz -shear strain are identical to the yz -shear strain results, which is expected due to the symmetry of the package. In Fig. 17, one can see that minimizing the CTE mismatch between die ($2.8 \times 10^{-6} \text{ K}^{-1}$) and lid reduces the amount of shear undergone by the TIM. At both extreme temperatures, the least amount of shear is observed by lid Mat. A ($3.1 \times 10^{-6} \text{ K}^{-1}$), followed by Mat. C ($7.7 \times 10^{-6} \text{ K}^{-1}$) and Mat. B ($8.3 \times 10^{-6} \text{ K}^{-1}$). With Cu lid ($17 \times 10^{-6} \text{ K}^{-1}$), the TIM displays the highest amount of shear. Considering the distance to the symmetry axis at the center of the package, for all lid materials the highest amount of shear is observed at the edges, as expected.

The variation of the maximum shear strain at the TIM as a function of the lid's Young's modulus and CTE is shown in Fig. 18. As for the maximum peel strain at the TIM, the maximum shear strain is mainly influenced by the CTE of the lid material, with higher CTE and lower E materials resulting in the highest shear, whereas lower CTE and lower E materials result in the lowest shear. For higher CTE materials, a small negative correlation between maximum shear strain and E can be seen, while for lower CTE materials a positive correlation between the maximum shear strain and E is observed. A positive correlation between the maximum shear strain and the CTE of the lid is observed for the studied range of E used in the vDoE.

5. Conclusion

A study concerning the usage of four different materials for the lid of a flip-chip package has been presented. For comparing the effects of using these materials, a FE numerical model was validated with experimental data obtained from warpage measurements at four different temperatures by using the projection moiré technique. For quantifying the influence of the different lid materials under study, three passive temperature cycles were applied to the numerical model and the warpage of the package was considered; the peel strain and plastic strain energy density at the bumps as well as the peel and shear strains at the TIM were evaluated.

Based on the results presented for the package warpage, as well as for the difference between maximum and minimum warpage (Δw) during the passive temperature cycles, it can be seen that there is a strong correlation between the CTE of the lid and the amount of warpage the package undergoes during temperature cycling. Packages with lid materials with low CTE (Mat. A, B and C) exhibit a higher Δw , while the package with Cu lid exhibits a much smaller Δw . The same is not observed when considering the Young's modulus of the lid materials, i.e., the stiffer material, Mat. A, results in the highest Δw , while Mat. B and C result in a very similar amount of Δw , although Mat. B is 1.8 times stiffer than Mat. C. Cu, on the other hand, has a very similar Young's modulus to Mat. B, but the resulting Δw is very different. Accordingly, it can be concluded that, when studying the influence of different materials for the package lid, the lid's most relevant material property influencing package warpage is the CTE.

Although lower Δw can be beneficial for board level interconnects, the results presented here show that lower Δw can result in higher peel strains at the bumps, which favors crack formation and propagation, as well as higher plastic strain energy density levels, which reduces the useful life of the flip-chip package and its reliability.

For the strains at the TIM, the results presented show that by having a CTE of the lid closer to the one of the Si die can be beneficial in terms of TIM reliability. By reducing the CTE-mismatch between die and lid, the strain state at the TIM decreases leading to reduced degradation of its thermal performance, once phenomena such as pump-out, interfacial delamination, adhesive and cohesive failures are minimized.

CRedit authorship contribution statement

Mateus Bagetti Jeronimo: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Visualization. **Jens Schindele:** Methodology. **Hubert Straub:** Conceptualization, Methodology, Supervision. **Przemyslaw Jakub Gromala:** Conceptualization, Methodology, Supervision. **Bernhard Wunderle:** Conceptualization, Supervision. **André Zimmermann:** Conceptualization, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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